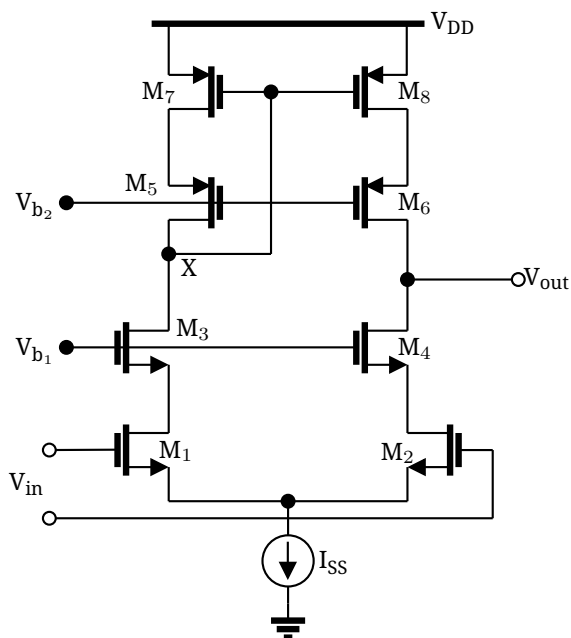


Use following parameters for this tutorial:

$\mu_n C_{ox} = 134.28 \mu A/V^2$, $\mu_p C_{ox} = 38.36 \mu A/V^2$, $\lambda_n = 0.1 V^{-1}$ for $L=0.5 \mu m$, $\lambda_p = 0.2 V^{-1}$ for $L=0.5 \mu m$, $V_{T0,n}=0.7 V$, $V_{T0,p}=-0.8 V$, $V_{DD}=3 V$

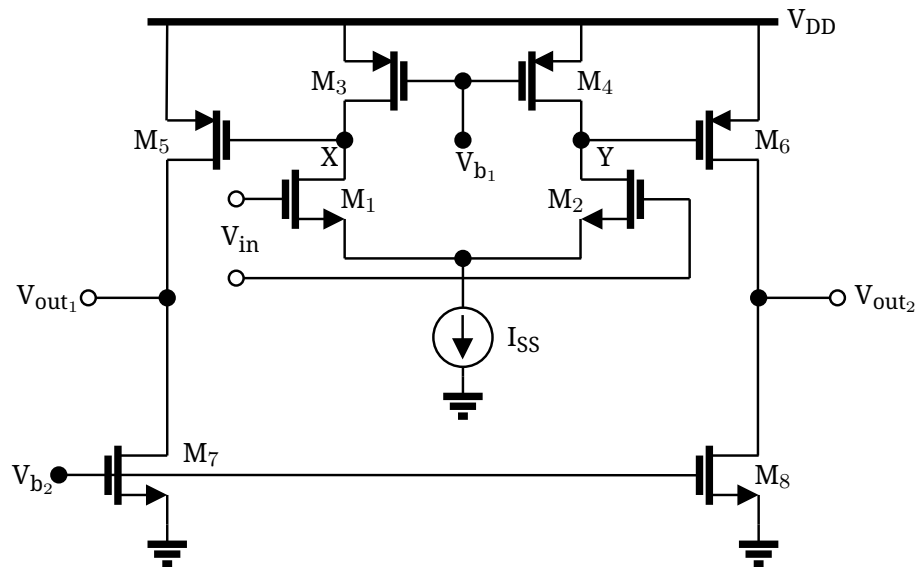
1. In the op amp of figure below, $(W/L)_{1-8} = 100/0.5$, $I_{SS} = 1 mA$, and $V_{b1} = 1.7 V$. Assume that $\gamma = 0$.

- What is the maximum allowable input CM level?
- What is V_X ?
- What is the maximum allowable output swing if the gate of M_2 is connected to the output?
- What is the acceptable range of V_{b2} ?



2. If in figure below, $(W/L)_{1-8} = 100/0.5$ and $I_{SS} = 1 mA$,

- What CM level must be established at the drains of M_3 and M_4 so that $I_{D5} = I_{D6} = 1 mA$? How does this constrain the maximum input CM level?
- With the choice made in part (a), calculate the overall voltage gain and the maximum output swing.



3. Suppose the circuit of figure below is designed with I_{SS} equal to 1 mA, $I_{D9} - I_{D12}$ equal to 0.5 mA, and $(W/L)_{9-12} = 100/0.5$.

- What CM level is required at X and Y ?
- If I_{SS} requires a minimum voltage of 400 mV, choose the minimum dimensions of $M_1 - M_8$ to allow a peak-to-peak swing of 200 mV at X and at Y .
- Calculate the overall voltage gain.

